

High-level design — keep the NVRs, share the hits

GUSIP sits beside existing cameras. Traffic, commissionerate, AMC, highway, railway, and the official Sentinel evaluation wall keep recording. We read streams, pull a still when needed, and publish a small event: who, where, when, plate, photo.

If a stolen Fortuner crosses five junctions in Ahmedabad and Gandhinagar, the operator should see one track — not five phone calls.

1. What we refuse to build

A statewide NVR (official Model 4). Copying ~80,000 continuous streams to one hall is a WAN and storage project, not an investigation project. Departmental VMS remains the system of record. GUSIP never writes PTZ or overwrites a local recording.

2. Picture of the system

```
Cameras / VMS / Sentinel wall      (video stays here)
    | read-only RTSP, ONVIF, vendor API, or Sentinel /stream
GUSIP adapter                      (ffmpeg still, health ping)
    | one event JSON
Bus (Redis now, Kafka later)
    | ANPR · YOLO · watchlist · coalesce
Control room                      (wall, yellow map pins, Investigate)
Stores: PostGIS cameras+tracks · Redis live+break-glass · MinIO stills
```

Companion drawing: architecture.svg. Same story as a left-to-right strip.

3. Two jobs the software must finish

Hit. Frame in → plate or person class → watchlist match → if that camera already has an open card, bump ×N. Else new alert + encrypted snapshot. WebSocket to the wall. Large pane switches. Target on the Paldi→Gandhinagar corridor: under 8 seconds.

Reconstruct. Officer picks a purpose (stolen vehicle, wanted person, evaluation...). Search will not run without it. Result is a GIS polyline of hops. Case folder can attach the latest alert. Export is blocked for operators.

4. Who sees what

Login	Sees	Cannot
operator	Statewide wall, ack alerts, search with purpose	CSV / case JSON
investigator	Above + watchlist write + export	Create users

coordinator (Ahmedabad)	Home district only	Other districts unless 5–120 min break-glass with FIR text
admin	Everything, including onboard and audit	—

5. PoC that is running vs later

	Now	Commissionerate / state
Gov data	31 Sentinel cameras, live proxy + Tesseract stills	Same adapter, more IDs
Own data	50 mixed RTSP / ONVIF / vendor-style cameras	Point adapters at real URLs
AI	YOLOv8n on CPU when installed; simulator shares the same event	Regional GPU, ByteTrack, Indian ANPR
Login	JWT, four demo users	Keycloak, MFA
Lookups	VAHAN / eGujCop / AFIS not called	After MoU + mTLS

6. Failure

If Surat's VMS dies, Surat cameras go grey. Ahmedabad and Sentinel keep playing. Worker reconnects. Break-glass grants die when Redis TTL says so, even if the coordinator forgets to click End.

Demo: <http://localhost:8080> · operator / GUSIP@ops2026 · plate GJ 01 ST 0001. Narrative docs in the repo: [architecture.md](#), [security.md](#), [scalability.md](#).